Unveil the Relationship Between Process and Design Embedded in the 3D Point Cloud using Unsupervised Learning

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Motivation

- **Investigate** : Variational patterns in 3D point cloud data related to geometric design and additive manufacturing processes.
- **Characterize**: Key (local) geometric features influencing point cloud accuracy and precision in 3D printed objects.

Objectives

- **Link process and design**: Reveal the relationship between geometric design and the accuracy and precision of the corresponding additive manufactured parts.
- **Local PGA pipeline**: Develop an unsupervised method to model the local variance and capture the geometric variational patterns.

Methods

- **Data Acquisition & Normalization**: Collect 3D scans; standardize or min–max scale coordinates
- **Spectral Clustering**: Build an affinity matrix, compute the Laplacian, then apply eigen-decomposition and k-means.
- **Label Alignment**: Use the Hungarian algorithm to match cluster labels across scans.
- **Local PGA**: Project clusters onto a tangent space; run PCA to extract principal geodesics and map them back.
- **Variance Validation**: Apply Kruskal–Wallis for variance checks; use ARI/NMI to assess clustering stability

Results

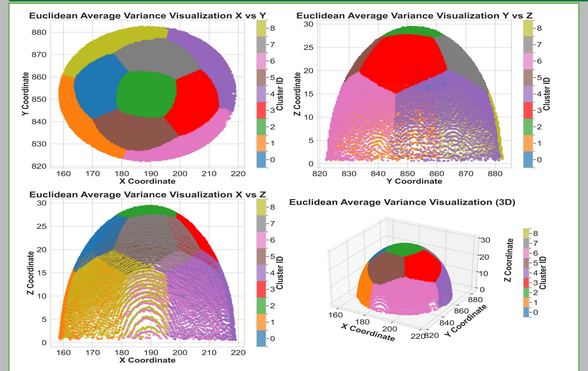


Figure 1: Average Variance for each cluster (Half-Ball).

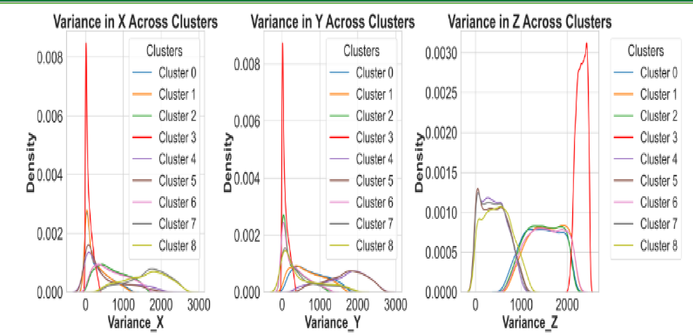


Figure 3: Variance Distribution Across Clusters for Half-Ball.

Variance analysis ($p < 0.05$) shows that curved regions have higher variability than flat areas (Table 1). The half-ball object has higher Z-axis variance, while the freeform object exhibits a more even distribution (Figures 1 and 3).

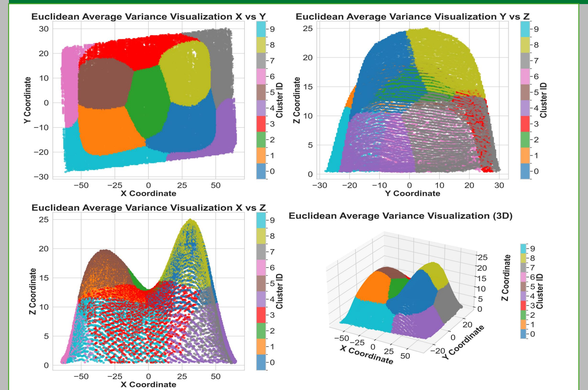


Figure 2: Average Variance for each cluster (Freeform).

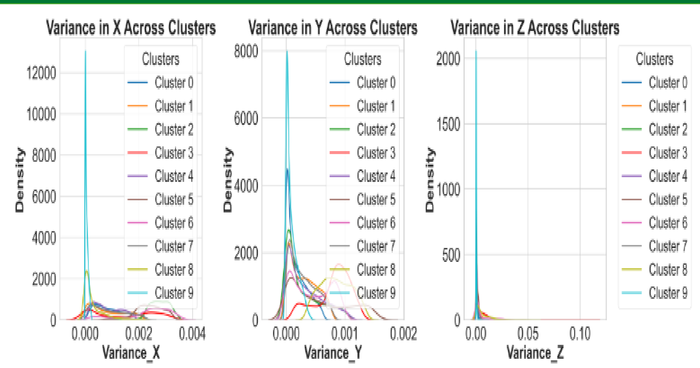


Figure 4: Variance Distribution Across Clusters for Freeform.

The pipeline consistently captures localized geometric variability (Table 2). Figures 2 and 4 highlight distinct variance patterns across clusters for both objects.

Table 1: Variance Analysis for Half-Ball and Freeform Measurements

Axis	Variance H-statistic ($p < 0.05$)	
Direction	Half-Ball (Mean ± SD)	Freeform (Mean ± SD)
X-axis	18,437 ± 500	17,698 ± 420
Y-axis	18,401 ± 400	17,833 ± 450
Z-axis	27,624 ± 60	27,142 ± 75

*All measurements significant at $p < 0.05$.

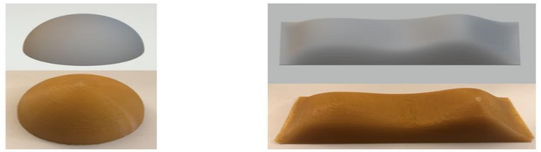
Table 2: Clustering Consistency Metrics Comparison (Overall, not axis-specific

Metric	Half-Ball (Mean ± SD)	Freeform (Mean ± SD)
ARI	0.85 ± 0.08	0.88 ± 0.06
NMI	0.88 ± 0.06	0.91 ± 0.05

†Consistent clustering performance observed.

Experiments Setup

- We evaluated scans of two 3D printed objects—a half-ball and a free-form structure—made with a LulzBot TAZ 3D printer using 2.85” polymer filament.



Half-Ball Freeform

Figure 5: The CAD models and physical objects considered in our study.

Discussion

- **Manufacturing signatures:** The PGA pipeline segments 3D point clouds into clusters that highlight specific process variations.
- **Geometry-process mapping:** Grouping geometric features links raw structure to manufacturing variations for targeted quality improvements..

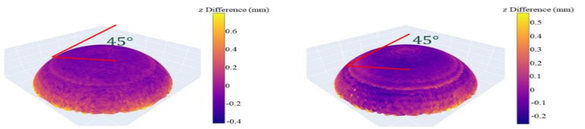


Figure 6: Visualization of the pointwise deviations with different layer thick-ness

Conclusion & Future work

- **Robust variance capture:** The Local PGA pipeline effectively segments point clouds, revealing localized geometric variability validated by statistical tests and consistent clustering.
- **Future integration:** Future work will link these variations to manufacturing quality and integrate defect detection for enhanced process control.